

1NF, 2NF, 3NF, bcNF, 4NF (Before & After examples rules)

First Normal Form (1NF) :-

Rules

- Each table must contain a primary key
- There are no repeating groups or no repeating attributes
- Each attribute must contain an atomic value

Before 1NF :-

Roll No	Name	Subject codes	Marks	Phone number
79	Piyush	24MA203, 24IT205	90, 100	9440226135, 9492528135

Subject codes and marks → multiple values

phone numbers → Not atomic

Repeating group exists

After 1NF :-

Student table	
RollNo (PK)	Name
79	Piyush

Marks table		
RollNo	Subject code	Marks
79	24MA203	90
79	24IT205	100

Primary key : (Roll No, Subject code)

Phone Number Table

RollNo	Phone Number
79	9440226135
79	9492528135

Composite key : (RollNo, Phone Number)

Second Normal Form (2NF) :-

Rules:-

- It must be in 1NF
- Every non key attribute is fully dependent on key attribute
- No partial dependency

Before 2NF:-

Roll No	Project ID	Name	Address	Project Name
79	24A	A	Vijayawada	LiFi

Primary key : (Roll No, Project ID)

Functional dependency : Roll No $\rightarrow$ Name, Address (partial dependency)
Project ID $\rightarrow$ Project Name (partial dependency)

After 2NF:-

Student details table

RollNo	Name	Address
79	A	Vijayawada

Project table

Project ID	Project Name
24A	LiFi

Work table

RollNo	Project ID
79	24A

Third Normal Form (3NF) :-

Rules

- It must be in 2NF
- No transitive dependency [Non key $\rightarrow$ Non key]

Before 3NF :-

Employee ID	Employee Name	Pincode	district	state
Z4A	Piyush	520003	NTR	AP

Primary key : (Employee ID) (Pincode)

After 3NF :-

Employee table

Employee ID	Employee Name	Pincode
Z4A	Piyush	520003

Functional dependency

Employee ID $\rightarrow$ Employee Name_{pin code}

Pincode $\rightarrow$ district, state

Address table

Pincode	District	State
520003	NTR	AP

Boyce - Codd Normal Form (BCNF) :-

Rules :-

- BCNF is not a normal form but it is higher version of 3NF
- BCNF is a 3.5 NF
- It is in 3NF
- For every functional dependency $F: x \rightarrow y$ x is superkey

Before BCNF:-

Relation : R (studID, Course Name, Teacher)

primary key: (studID, Course Name)

studID	Course Name	Teacher
A4	DBMS	Ramesh
A5	JAVA	Ashok

After BCNF:-

student teacher table

studID	Teacher
A4	Ramesh
A5	Ashok

Teacher Course Table

Teacher	Course Name
Ramesh	DBMS
Ashok	JAVA

Functional dependency

(studID, Course Name) $\rightarrow$ Teacher

Teacher $\rightarrow$ Course Name

Fourth Normal Form (4NF):-

Rules:-

$\rightarrow$ It should be in 3NF

$\rightarrow$ Not multivalued dependency

Before 4NF:-

Relation: R (Bike model, Year, color)

Bike model	Year	color
M2011	2008	White
M2001	2008	Black

multivalued dependency

Bike model $\rightarrow$ Year,

Bike model $\rightarrow$ color

After 4NF:-

Bike-year table

Bike model	Year
M2011	2008
M2001	2008

Bike colour

Bike model	color
M2011	white
M2001	Black

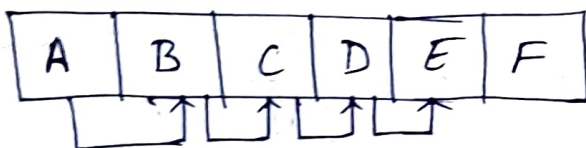
Identify Candidate key and find is it in 2nf or 3nf give examples.

A candidate key in DBMS is a minimal set of attributes that can uniquely identify each record in table

Ex:-

① $R(A, B, C, D, E, F)$

FD: $A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E$



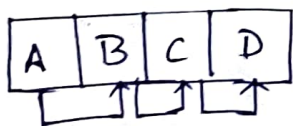
$$(AF)^+ = \{A, F, B, C, D, E\}$$

AF is candidate key of relation R

$\therefore$ only 1nf (not in 2nf or 3nf)

② $R(A, B, C, D)$

FD: $A \rightarrow D, B \rightarrow C, C \rightarrow D$



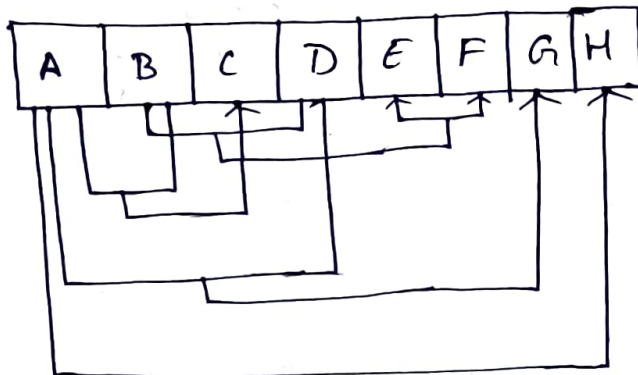
$$(A)^+ = \{A, B, C, D\}$$

candidate key = A

$\therefore$ it is in 2nf but not in 3nf

③ $R(A, B, C, D, E, F, G, H)$

FD: $AB \rightarrow C$, $BD \rightarrow EF$, $AD \rightarrow G$, $A \rightarrow H$



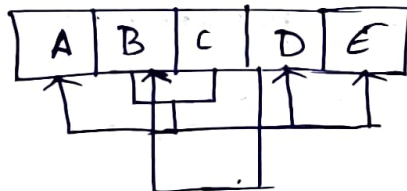
$$(ABD)^+ = \{A, B, D, C, E, F, G, H\}$$

Candidate key = ABD

$\therefore$ only in 1NF (not in 2nf or 3nf)

④ $R(A, B, C, D, E)$

FD: $BC \rightarrow ADE$, $D \rightarrow B$



$$C^+ = \{C, x\}$$

$$(CA)^+ = \{C, A, x\}$$

$$(CB)^+ = \{C, B, A, D, E\} \checkmark$$

$$(CD)^+ = \{C, D, B, A, E\} \checkmark$$

$$(CE)^+ = \{C, E, x\}$$

CD is the candidate key of particular relation R

$\therefore$ D CD determines B $\rightarrow$ partial dependency

$\therefore$ only in 1NF (not in 2nf or 3nf)